Hyperspectral Image Reconstruction of SD-CASSI Based on Nonlocal Low-Rank Tensor Prior

Xiaorui Yin, Lijuan Su, *Member, IEEE*, Xin Chen, Hejian Liu, Qiangqiang Yan, and Yan Yuan

***Abstract*— In single disperser coded aperture snapshot spectral imaging (SD-CASSI) systems, many methods have been developed to reconstruct hyperspectral images (HSIs) from compressed measurements. Among these, deep learning (DL)-based methods have stood out, relying on powerful DL networks. However, the solidified structure of DL-based methods limits their adaptability. Moreover, they are often based on a model that neglects the dispersion process and instead emphasizes the encoding-compression process. Furthermore, research on optimization-based methods designed especially for SD-CASSI is lacking. In this article, we propose a comprehensive two-step projection imaging model for SD-CASSI that includes both spectral shearing projection and encoding-compression projection. Based on this model, we derive a tensor-based optimization framework that incorporates with the nonlocal low-rank tensor (NLRT) prior. In particular, NLRT extracts inherent spatial structural information from the measurements and employs it to guide the clustering of spatial–spectral similar HSI blocks. A CANDECOMP/PARAFAC (CP) low-rank regularizer is introduced to constrain the low-rank property of HSI block clusters. After that, we develop a solution framework based on the alternating direction method of multiplier (ADMM) approach. Comprehensive experiments demonstrate that our NLRT method outperforms state-of-the-art methods in terms of flexibility and performance. The source code and data of this article are publicly available at https://github.com/sdnjyxr/NLRT.**

***Index Terms*— Hyperspectral images (HSIs), nonlocal low-rank tensor (NLRT), single disperser coded aperture snapshot spectral imaging (SD-CASSI), two-step projection model.**

I. Introduction

THE snapshot hyperspectral imaging technique can acquire 3-D hyperspectral images (HSIs) composed of two spatial dimensions and one spectral dimension within a single exposure time. HSIs contain spatial radiation information of hundreds of contiguous and narrow spectral bands associated with various targets. Therefore, HSIs have attracted great attention in various fields, such as remote sensing [1], [2], medical diagnostics [3], military surveillance [4], target tracking [5], and so on. Commonly employed direct modulation snapshot hyperspectral imaging techniques, such as the image mapping spectrometer (IMS) [6] and the multispectral light field imager (MSLFI) [7], [8], divide the 3-D HSI datacube and redistribute it onto different positions of a 2-D sensor. However, limited by the pixel number of the sensor, there is an inevitable tradeoff between the spatial resolution and spectral resolution of the obtained HSIs.

Compressive sensing (CS) is a type of imaging and processing technology that enables the reconstruction of the original image with fewer measurements. Inspired by this theory, several spectral snapshot compressive imaging (SSCI) systems have been proposed to acquire HSIs [9], [10], [11], [12], [13]. Specifically, the coded aperture snapshot spectral imaging (CASSI) system is a typical SSCI system that modulates a 3-D HSI into a 2-D distribution after spatial coding, spectral dispersing, and compressing. Different from direct modulation snapshot spectral imaging technologies, a full-resolution HSI can be obtained after applying a reconstruction algorithm to a CASSI image. According to the number of dispersive elements, CASSI systems can be classified into the dual-disperser CASSI (DD-CASSI) system [9] and the single-disperser CASSI (SD-CASSI) system [14]. Compared with SD-CASSI, DD-CASSI has a more complex system structure. Its measurement is a summation of encoded images of all spectral bands. On the other hand, the measurement of SD-CASSI is a summation of a series of images that have been shear-coded. As illustrated in Fig. 1(a), a typical SD-CASSI system consists of a dispersive element, an aperture mask, an objective lens, and two relay lenses. As shown in Fig. 1(b), the shearing effects disrupt the spatial correspondence between different spectral bands, which is detrimental to the HSI reconstruction of SD-CASSI. The main challenge in the application of the SD-CASSI systems lies in the accurate reconstruction of 3-D HSIs from 2-D compressive measurement images.

Obviously, reconstructing a 3-D HSI from a 2-D compressed image is an ill-posed problem. In the past decades, optimization-based methods and deep learning (DL)-based methods have been proposed for solving original HSI. Early optimization-based methods were matrix-based, such as two-step iterative shrinkage/thresholding (TwIST) [15] and gradient projection for sparse reconstruction (GPSR) [16], which unfolded the 3-D HSI cube into a 1-D vector, destroying

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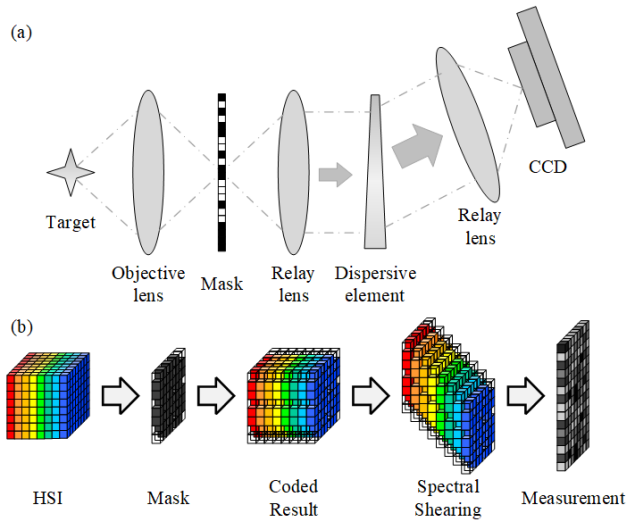


Fig. 1. (a) Structure of the SD-CASSI system. The target is captured by the objective lens, spatially sheared by the mask, spectrally coded by the dispersive element, and imaged on the CCD. The mask, the dispersive element, and the CCD lie within the conjugate image plane of the scene. (b) Process of HSI compressing in SD-CASSI system.

the inherent multidimensional structure of HSIs, yielding unsatisfactory results. Moreover, generalized alternating projection with total variation (GAP-TV) [17] and decompress snapshot compressive imaging (DeSCI) [18] considered 2-D spatial correlations, but the reconstruction quality remained notably deficient.

In recent years, DL-based methods, which can be categorized into fully learned methods and model-based methods, have been proposed for CASSI to achieve better results. The fully learned methods, such as λ-net [19], coarse-to-fine sparse transformer (CST) [20], mask-guided spectral-wise transformer (MST) [21], and bidirectional recurrent neural networks with adversarial training (BIRNAT) [22], can deliver satisfactory results with low time cost, but they lack interpretability. On the other hand, the model-based methods, such as GAP-Net [23], ADMM-Net [24], and degradation-aware unfolding half-shuffle transformer (DAUHST) [25], offer more interpretability than fully learned methods and lead to SOTA performance. However, DL-based methods inevitably have the following disadvantages. First, it is challenging to create a training dataset that matches the sensing matrix of the testing phase in a practical CASSI system. Second, since DL-based methods are trained, verified, and tested on datasets with limited scenes, they may not achieve sufficient performance when applied to scenes beyond the utilized datasets. Third, without retraining, the solid structure of DL-based methods renders them unsuitable for direct application in practical CASSI systems with distinct imaging parameters, such as spatial coding patterns, spatial and spectral dimension sizes, and compression ratios.

Recently, tensor analysis methods for HSIs are widely applied in HSI processing tasks, including HSI completion [26], HSI denoising [27], and HSI super-resolution [28]. Tensor rank is a classical parameter that describes the multidimensional structure of tensors. CANDECOMP/PARAFAC (CP) decomposition [29] and Tucker decomposition [30] are the most commonly used tensor low-rank decomposition strategies. Tensor models fully exploit the inherent multilinear structure in HSIs to provide better understanding and richer information [31]. Nonlocal meets global (NGmeet) [32], [33], as a unified tensor iteration paradigm for HSI restoration, integrates spatial nonlocal similarity prior and global spectral low-rank prior of HSI, demonstrating excellent performance in HSI denoising, reconstruction, and inpainting. However, NGmeet separates the spatial correlation and spectral low-rankness of HSI for discussion, overlooking their intrinsic connection. In SD-CASSI systems, the shearing effect and modulation of coded patterns significantly degrade the quality of spatial structure. The classical SD-CASSI model was not conducive to analyzing the correlation between HSI spatial and spectral dimensions. Despite the fact that SD-CASSI systems are more compact and user-friendly, it is worth noting that existing optimization-based methods rarely consider HSI reconstruction within the context of SD-CASSI systems.

In this article, we establish a two-step projection imaging model for SD-CASSI and introduce a tensor-based optimization framework named nonlocal low-rank tensor (NLRT) for HSI reconstruction. To address the aforementioned degradation of the measured images, we restore the measurement to a smooth grayscale image by introducing a total variation (TV) prior. Natural images are inherently self-similar, thereby containing numerous nonlocal similar patches. The grayscale image shares a similar spatial structure with the target HSI [34], and thus can be used to extract the self-similar information of the target HSI. To be specific, a number of patches are initially created by partitioning the grayscale image. For each patch, the similar patches extracted from its surrounding region are grouped into a cluster. Subsequently, the nonlocal HSI blocks associated with each spatially correlated cluster are collectively formed into a 4-D low-rank tensor comprising two spatial modes, one spectral mode, and one similarity mode. In order to constrain the low-rank property, a CP-norm regularization term is introduced into the optimization problem. The reconstructed HSI is obtained using the alternating direction method of multipliers (ADMM). The main contributions can be summarized as follows.

1) We propose a tensor-based imaging model for SD-CASSI, comprising a two-step linear projection process. The first process is a spectral shearing projection, and the second process is an encoding-compression projection. This model provides a comprehensive description of the HSI acquisition process in the CASSI system. The model complexity is reduced by decomposing the spatial–spectral compression process into two simple linear projection procedures.
2) An image restoration method is proposed to restore spatial information from the measurement image of SD-CASSI. We introduce a TV regularizer to generate an initial HSI and the restored image is calculated as the average of each spectral 2-D slice of the initial estimated 3-D tensor. The reconstructed image effectively retains spatial structural details, thereby enhancing the performance of NLRT.

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3) The NLRT prior is introduced into the HSI reconstruction problem of SD-CASSI. The target HSIs are obtained by using the ADMM approach to solve the reconstruction problem. Experiments demonstrate that the performance of the proposed NLRT method outperforms SOTA methods.

This article is organized as follows. The reconstruction methods of CASSI are reviewed in Section II. The mathematical notations and definitions are elaborated in Section III. In Section IV, based on the physical imaging mechanism of SD-CASSI, a two-step projection tensor-based model is established. Section V describes the framework of the proposed NLRT method, and the solving algorithm based on ADMM is provided. Section VI presents the experimental findings and discussions. Finally, the conclusion is drawn in Section VII.

II. Related Works

The CASSI technique exploits the CS theory to project a 3-D HSI into a 2-D measurement in a single exposure. Undoubtedly, restoring a 3-D HSI from a 2-D measurement image is an ill-posed problem. In this section, we review the HSI reconstruction methods for CASSI. These methods can be classified into two categories: optimization-based methods and DL-based methods.

A. Optimization-Based Methods

Optimization-based methods reconstruct HSIs by solving an optimization problem with different priors. In the past decade, a variety of handcrafted priors, including sparsity [11], [16], TV [17], [35], [36], nonlocal similarity [18], [37], [38] and Gaussian mixture model (GMM) based prior [39], [40], have been employed in the reconstruction methods. The optimization-based methods can be divided into two categories: matrix-based methods and tensor-based methods.

Various matrix-based methods have been used for the HSI reconstruction of CASSI. TwIST [15] and GPSR [16] utilize the sparsity of signals to reconstruct HSI data. However, due to the high complexity of this algorithm, there is a considerable time and memory overhead. GAP-TV [17] is a typical reconstruction method, which makes use of the sparsity and TV properties of HSIs. Although it may deliver results within seconds, the reconstruction quality is unsatisfactory. DeSCI [18] enhanced the output of GAP-TV by utilizing the nonlocal self-similarity characteristics of HSIs, and simultaneously conducting nonlocal similar patch search in a 3-D window composed of spatial and spectral dimensions. However, it ignored the intrinsic dispersion patterns of the system. Recently, many methods based on deep priors have been proposed. These methods combined optimization approaches for HSI reconstruction with pretrained or untrained deep networks as regularization terms. In order to achieve better results than manually defined priors, the plug-and-play deep image prior (PnP-DIP) [41] method proposed a self-supervised neural network framework, which incorporates the PnP-DIP and directly learns priors from measurement images using a deep decoder. TV-FFDNet [42] significantly improved the reconstruction effectiveness of HSI by combining the deep denoising prior (DDP) and TV prior within the PnP framework.

An HSI can be naturally represented as a 3-D tensor, and tensor-based methods have demonstrated the ability to effectively maintain the inherent spatial–spectral self-similarity characteristics of HSIs. NGmeet [33] first learned a global spectral low-rank orthogonal basis and subsequently exploited the spatial nonlocal similarity of the projected HSI on this basis. LR2DN [43] incorporated unsupervised DIP and pretrained DDP into the PnP framework, effectively capturing spectral correlation and depth spatial features in HSI. It demonstrated excellent generalization capabilities and improved reconstruction accuracy. By integrating model-driven low-rank representation with self-supervised DL, LRSDN [44], a SOTA self-supervised method, introduced a new HSI reconstruction framework and enhanced reconstruction quality and adaptability. It is noteworthy that none of these methods have taken into account both the problem of structural distortion caused by shear degradation in the SD-CASSI system and the 3-D structural characteristics of HSI.

B. DL-Based Methods

In recent years, DL-based methods have stood out for the HSI reconstruction task within the CASSI system, relying on powerful DL networks. They can deliver excellent results within a few seconds. There are two categories of DL-based algorithms [45]: fully learned methods and model-based methods.

Fully learned methods take advantage of their powerful learning capability to directly recover HSIs from measurement images. MST [21] first applied the transformer to the HSIs reconstruction tasks, and introduced a spectral-wise multihead self-Attention module. CST [20] embedded HSIs sparsity into the network and obtained great results. BIRNAT [22] considered the sequential correlation in HSIs, completing the HSI reconstruction with both high reconstruction quality and fast inference time. However, these methods lack interpretability.

Model-based methods reconstruct HSIs data based on mathematical and physical models of CASSI. GAP-Net [23] unfolded the GAP algorithm and employed a trained convolutional neural network for the estimation of the desired signal. ADMM-Net [24] was designed on the basis of a standard tensor ADMM algorithm, and unfolded its inference iterations into a layer-wise structure. DAUHST [25] estimated parameters from the compressed image and a physical mask and subsequently utilized these parameters to control each iteration.

III. Notations and Preliminaries

Throughout the article, scalars are denoted as italic Greek letters and regular letters (e.g., β, a). Vectors are denoted as bold lowercase letters (e.g., $\mathbf{x}$, $\mathbf{y}$). Matrices are denoted as bold uppercase letters (e.g., $\mathbf{A}$, $\mathbf{M}$). Tensors are denoted as calligraphic bold uppercase letters (e.g., $\boldsymbol{\mathcal{X}}$, $\boldsymbol{\mathcal{L}}$). Linear operators are denoted as calligraphic uppercase letters (e.g., $\mathcal{H}$, $\mathcal{A}$), and $\mathcal{A}(\boldsymbol{\mathcal{X}})$ denotes the result of applying the linear

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operator $\mathcal{A}$ to the tensor $\boldsymbol{\mathcal{X}}$. The adjoint of $\mathcal{A}$ is denoted by $\mathcal{A}^\dagger$.

The field of real number is denoted as $\mathbb{R}$. An N-order tensor can be denoted by $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{n_1 \times n_2 \cdots \times n_N}$. The $(i_1, i_2, \ldots, i_N)$-element of tensor $\boldsymbol{\mathcal{X}}$ is denoted as $x_{i_1 i_2, \ldots, i_N}$ or $\boldsymbol{\mathcal{X}}(i_1, i_2, \ldots, i_N)$. The Frobenius norm of tensor $\boldsymbol{\mathcal{X}}$ is defined as $\|\boldsymbol{\mathcal{X}}\|_F = (\sum_{i_1,\ldots,i_N} |x_{i_1,\ldots,i_N}|^2)^{1/2}$.

For a three-order tensor $\boldsymbol{\mathcal{X}}$, the ith frontal slice is denoted as $\boldsymbol{\mathcal{X}}(:, :, i)$ or $\boldsymbol{\mathcal{X}}^{(i)}$. Two operators, *unfold* and *fold*, are, respectively, defined as

$$\text{unfold}(\boldsymbol{\mathcal{X}}) = \begin{bmatrix} \boldsymbol{\mathcal{X}}^{(1)} \\ \boldsymbol{\mathcal{X}}^{(2)} \\ \vdots \\ \boldsymbol{\mathcal{X}}^{(n_3)} \end{bmatrix} \tag{1}$$

$$\text{fold}(\text{unfold}(\boldsymbol{\mathcal{X}})) = \boldsymbol{\mathcal{X}}. \tag{2}$$

Definition 1 (CP Approximation): For a N-order tensor $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{n_1 \times n_2 \times \cdots \times n_N}$, the CP approximation of $\boldsymbol{\mathcal{X}}$ is denoted as

$$\begin{aligned} \mathcal{F}_{\text{CP}}(\boldsymbol{\mathcal{X}}, r) := \underset{\boldsymbol{\mathcal{Z}}}{\text{argmin}} \, \|\boldsymbol{\mathcal{Z}} - \boldsymbol{\mathcal{X}}\|_F^2 \\ \text{s.t. } \boldsymbol{\mathcal{Z}} = [[\mathbf{Z}_1, \ldots, \mathbf{Z}_B]] \end{aligned} \tag{3}$$

where $[[\cdots]]$ is a shorthand notation, also termed as the Kruskal operator, the expression $\boldsymbol{\mathcal{Z}} = [[\mathbf{Z}_1, \ldots, \mathbf{Z}_B]]$ indicates that the tensor $\boldsymbol{\mathcal{Z}}$ obeys the CP-factorization [46], [47] with factors $\mathbf{Z}_i \in \mathbb{R}^{n_i \times r}, i = 1, 2, \ldots, N$ for a CP-rank r.

Definition 2 (Tensor Inner Product): The inner product of $\boldsymbol{\mathcal{X}}$ and $\boldsymbol{\mathcal{M}} \in \mathbb{R}^{n_1 \times n_2 \times \cdots \times n_N}$ is denoted as

$$\langle \boldsymbol{\mathcal{X}}, \boldsymbol{\mathcal{M}} \rangle = \sum_{i_1,\ldots,i_N} \boldsymbol{\mathcal{X}}(i_1, \ldots, i_N) \cdot \boldsymbol{\mathcal{M}}(i_1, \ldots, i_N). \tag{4}$$

Additional notation will be defined in relevant sections if necessary.

IV. SD-CASSI Model

In this section, based on the physical imaging characteristics of SD-CASSI, we present a tensor-based mathematical model that incorporates two linear projections. In comparison with previous imaging models, our model divides a complex spatial–spectral modulation process into two basic linear projecting procedures, which reduces the complexity of the subsequent optimization problem-solving.

A. Physical Imaging Mechanism of SD-CASSI

A point on the target HSI cube is denoted as $\boldsymbol{\mathcal{X}}_0(i, j, \lambda)$, where i and j are spatial mode indices, while λ is spectral mode index. After captured by the objective lens and modulated by the mask, $\boldsymbol{\mathcal{X}}_1(i, j, \lambda)$ is given by

$$\boldsymbol{\mathcal{X}}_1(i, j, \lambda) = \mathbf{T}(i, j)\boldsymbol{\mathcal{X}}_0(i, j, \lambda) \tag{5}$$

where $\mathbf{T}(i, j)$ is the transmission pattern of the mask. Subsequently, the prism introduces spectral dispersion. Before the CCD, the $\boldsymbol{\mathcal{Y}}(i, j, \lambda)$ is given by

$$\begin{aligned} \boldsymbol{\mathcal{Y}}(i, j, \lambda) &= \boldsymbol{\mathcal{X}}_1(i + \alpha(\lambda - \lambda_c), j, \lambda) \\ &= \mathbf{T}(i + \alpha(\lambda - \lambda_c), j)\boldsymbol{\mathcal{X}}_0(i + \alpha(\lambda - \lambda_c), j, \lambda) \end{aligned} \tag{6}$$

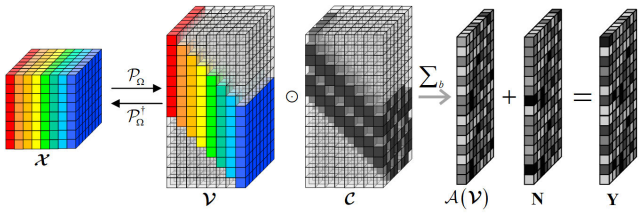


Fig. 2. Visual representation of the spectral shearing projection process and the encoding-compression projection process.

where α denotes a dispersive element with linear dispersion and λ_c denotes the center wavelength. And then $\boldsymbol{\mathcal{Y}}(i, j, \lambda)$ is integrated along with spectral mode by the sensor. The measurement $\mathbf{Y}(i, j)$ is denoted as

$$\begin{aligned} \mathbf{Y}(i, j) &= \sum_\lambda \boldsymbol{\mathcal{Y}}(i, j, \lambda) \\ &= \sum_\lambda \boldsymbol{\mathcal{C}}(i, j, \lambda)\boldsymbol{\mathcal{X}}(i, j, \lambda) \end{aligned} \tag{7}$$

where $\boldsymbol{\mathcal{C}}$ and $\boldsymbol{\mathcal{X}}$ can be regarded as 3-D spectral sheared tensors.

B. Tensor-Based Two-Step Projection Model

In SD-CASSI, we denote the measurement as $\mathbf{Y} \in \mathbb{R}^{(m+\alpha(B-1)) \times n}$, where m and n denote the height and width of HSI, B is the number of spectral channels, α denotes the dispersion shift step. The underlying 3-D HSI cube is denoted as $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{m \times n \times B}$. Different from the previous imaging model, we split the process of compression imaging into a spectral shearing projection and an encoding-compression projection and propose a two-step projection model.

As shown in Fig. 2, the underlying HSI cube $\boldsymbol{\mathcal{X}}$ is first dispersed to obtain the 3-D spectral sheared tensor $\boldsymbol{\mathcal{V}}$, which is the spectral shearing projection process. The encoding process can be regarded as the tensor $\boldsymbol{\mathcal{V}}$ undergoing a Hadamard product with an equivalent mask tensor $\boldsymbol{\mathcal{C}}$, which is generated from transmission pattern matrix $\mathbf{T}$. The compression process involves summing the encoding result along the spectral dimension to obtain the encoded compressed result $\mathcal{A}(\boldsymbol{\mathcal{V}})$. The measurement matrix $\mathbf{Y}$ is then generated by combining the compressed results with the detection noise matrix $\mathbf{N}$.

The modulation and compression process can be written as follows:

$$\mathbf{Y} = \mathcal{A}(\mathcal{P}_\Omega(\boldsymbol{\mathcal{X}})) + \mathbf{N} \tag{8}$$

where $\mathbf{N} \in \mathbb{R}^{(m+\alpha(B-1)) \times n}$ denote the additive noise, $\mathcal{P}_\Omega(\cdot) : \mathbb{R}^{m \times n \times B} \to \mathbb{R}^{(m+\alpha(B-1)) \times n \times B}$ is the spectral shearing projection operator, which can be expressed as

$$(\mathcal{P}_\Omega(\boldsymbol{\mathcal{X}}))_{ijk} = \begin{cases} \boldsymbol{\mathcal{X}}(i - \alpha(k-1), j, k), & i, j, k \in \Omega \\ 0, & i, j, k \notin \Omega \end{cases} \tag{9}$$

where Ω denotes the set of nonzero element indices of the dispersed HSI cube. Its adjoint operator $\mathcal{P}_\Omega^\dagger(\cdot) : \mathbb{R}^{(m+\alpha(B-1)) \times n \times B} \to \mathbb{R}^{m \times n \times B}$ can be expressed as

$$\left(\mathcal{P}_\Omega^\dagger(\boldsymbol{\mathcal{V}})\right)_{ijk} = \boldsymbol{\mathcal{V}}(i + \alpha(k-1), j, k) \tag{10}$$

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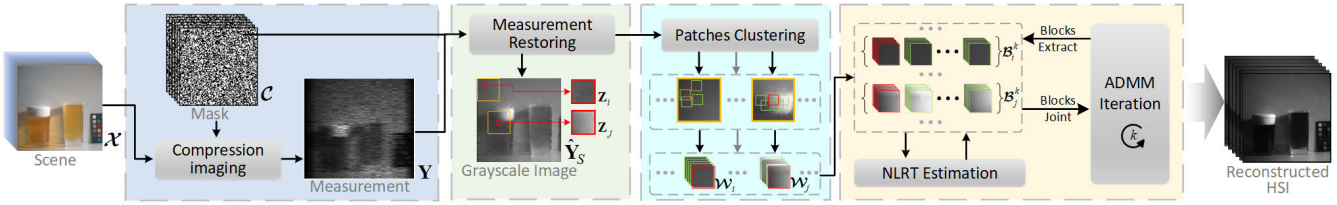


Fig. 3. Illustration of NLRT framework. Blue box: the process of compression imaging. Green box: the process of restoring the measurement image. Cyan box: the process of clustering similar image patches. Yellow box: the reconstruction algorithm based on ADMM and NLRT prior, with separate reconstruction of each cluster (obtained from the previous step).

where $\boldsymbol{\mathcal{V}} \in \mathbb{R}^{(m+\alpha(B-1))\times n\times B}$ denotes a dispersed HSI cube.

Regarding the operator $\mathcal{P}_\Omega$, the following properties obviously hold:

1) $\|\boldsymbol{\mathcal{X}}\|_F^2 = \|\mathcal{P}_\Omega(\boldsymbol{\mathcal{X}})\|_F^2$
2) $\|\boldsymbol{\mathcal{V}}\|_F^2 = \|\mathcal{P}_\Omega^\dagger(\boldsymbol{\mathcal{V}})\|_F^2$, when $\boldsymbol{\mathcal{V}}(i,j,k)=0,\ i,j,k\notin\Omega$
3) $\mathcal{P}_\Omega^\dagger(\mathcal{P}_\Omega(\boldsymbol{\mathcal{X}})) = \boldsymbol{\mathcal{X}}$
4) $\mathcal{P}_\Omega(\mathcal{P}_\Omega^\dagger(\boldsymbol{\mathcal{V}})) = \boldsymbol{\mathcal{V}}$, when $\boldsymbol{\mathcal{V}}(i,j,k)=0,\ i,j,k\notin\Omega$.

In addition, $\mathcal{A}(\cdot):\mathbb{R}^{(m+\alpha(B-1))\times n\times B}\to\mathbb{R}^{(m+\alpha(B-1))\times n}$ denotes the encoding-compression projection which is expressed as

$$\mathcal{A}(\boldsymbol{\mathcal{V}}) = \sum_b \boldsymbol{\mathcal{V}}^{(b)} \odot \boldsymbol{\mathcal{C}}^{(b)} \tag{11}$$

where $\odot$ denotes the Hadamard (element-wise) production, $\boldsymbol{\mathcal{C}}$ is generated by the transmission pattern

$$\boldsymbol{\mathcal{C}}(i,j,k) = \begin{cases} \mathbf{T}(i-\alpha(k-1),j), & i,j,k\in\Omega \\ 0, & i,j,k\notin\Omega. \end{cases} \tag{12}$$

The adjoint operator $\mathcal{A}^\dagger(\cdot):\mathbb{R}^{(m+\alpha(B-1))\times n}\to\mathbb{R}^{(m+\alpha(B-1))\times n\times B}$ can be expressed as

$$\mathcal{A}^\dagger(\mathbf{Y}) = \text{fold}\left(\begin{bmatrix} \mathbf{Y}\odot\boldsymbol{\mathcal{C}}^{(1)} \\ \mathbf{Y}\odot\boldsymbol{\mathcal{C}}^{(2)} \\ \vdots \\ \mathbf{Y}\odot\boldsymbol{\mathcal{C}}^{(B)} \end{bmatrix}\right) \tag{13}$$

where $\mathcal{A}(\mathcal{A}^\dagger(\mathbf{Y})) = \mathbf{Y}\odot\sum_b(\boldsymbol{\mathcal{C}}^{(b)}\odot\boldsymbol{\mathcal{C}}^{(b)}) = \mathbf{Y}\odot\mathcal{A}(\boldsymbol{\mathcal{C}})$, and $\mathcal{A}(\boldsymbol{\mathcal{C}})$ can be precomputed.

V. Proposed NLRT Approach

In this section, we present an NLRT framework for HSI reconstruction of SD-CASSI. As illustrated in Fig. 3, the framework includes three parts, which are elaborated in the following section. In Section V-A, a measurement restoration method is performed to obtain a grayscale image from the severely degraded image of the SD-CASSI measurement. In Section V-B, we leverage patch-based matching of the restored image with its surrounding patches to generate clusters with similar spatial–spectral structures. In addition, we establish an optimization model incorporating the CP-norm for further processing. In Section V-C, the ADMM method is utilized to reconstruct the desired HSIs, and the solutions for each subproblem are derived within this framework.

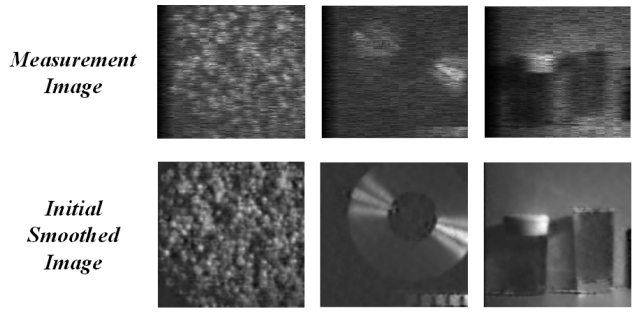


Fig. 4. Visualization comparison between measurement images observed by SD-CASSI system and grayscale estimation images. It can be seen that the effects of encoding and dispersion have been successfully removed.

A. Measurement Image Restoration

A HSI has a high self-correlation in the spatial domain. The inherent spatial structure of the HSI is still preserved in the 2-D measurement image. Nonetheless, for SD-CASSI, the spatial structure of the measurement is severely deteriorated due to the effects of spatial code modulation and spectral dispersion. In order to extract the spatial auxiliary information embedded within the measurement to assist with the HSI reconstruction, we proposed a measurement-restored algorithm to generate a smooth grayscale image from the measurement. Concretely, a TV regularizer is introduced, and the ADMM is employed to estimate a preliminary HSI. The ultimate restored image is obtained by averaging along the spectral mode. Visual representations comparing the original SD-CASSI measurements with the estimated smooth grayscale images are presented in Fig. 4. Further details regarding the estimation process are elaborated below.

Initially, a smooth estimation $\hat{\mathbf{Y}}_S$ can be regarded as the average of each spectral slice of a 3-D preliminary tensor $\boldsymbol{\mathcal{T}} \in \mathbb{R}^{m\times n\times B}$. In order to obtain a smooth solution, a TV regularization term is introduced. The formulation of the model for the preliminary tensor estimation $\boldsymbol{\mathcal{T}}$ is as follows:

$$\underset{\boldsymbol{\mathcal{T}},\boldsymbol{\mathcal{S}}}{\text{argmin}} \frac{1}{2}\|\mathcal{A}(\boldsymbol{\mathcal{S}})-\mathbf{Y}\|_F^2 + \rho\|\boldsymbol{\mathcal{T}}\|_{\text{TV}}, \quad \text{s.t. } \boldsymbol{\mathcal{S}} = \mathcal{P}_\Omega(\boldsymbol{\mathcal{T}}) \tag{14}$$

where ρ is a regularization parameter. The augmented Lagrangian function for the above problem can be written as

$$\min_{\boldsymbol{\mathcal{T}},\boldsymbol{\mathcal{S}},\boldsymbol{\mathcal{E}}} \mathcal{L}(\boldsymbol{\mathcal{T}},\boldsymbol{\mathcal{S}},\boldsymbol{\mathcal{E}}) := \frac{1}{2}\|\mathcal{A}(\boldsymbol{\mathcal{S}})-\mathbf{Y}\|_F^2 + \rho\|\boldsymbol{\mathcal{T}}\|_{\text{TV}} + \frac{\omega}{2}\left\|\boldsymbol{\mathcal{S}}-\mathcal{P}_\Omega(\boldsymbol{\mathcal{T}})-\frac{1}{\omega}\boldsymbol{\mathcal{E}}\right\|_F^2 \tag{15}$$

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Algorithm 1 Smooth Grayscale Estimation

Input: tensor $\mathcal{C}$, matrix $\mathbf{Y}$, regularization parameter $\rho > 0$, penalty parameter $\omega > 0$.

Initialize: $\mathcal{T}^1, \mathcal{S}^1, \mathcal{E}^1, max_iters$.

1: **for** k:=1 to max_iters **do**
2: Update $\mathcal{S}^{k+1}$ using Eq. (20);
3: Update $\mathcal{T}^{k+1}$ using Eq. (21);
4: Update $\mathcal{E}^{k+1}$ using Eq. (18);
5: **end for**

Output: The smooth estimation result with Eq. (22).

where $\mathcal{E}$ represents a Lagrangian multiplier and ω is a penalty parameter.

The ADMM method is used to solve the above problem, then (15) can be translated into three subproblems

$$\mathcal{S}^{k+1} = \underset{\mathcal{S}}{\operatorname{argmin}} \frac{1}{2}\|\mathcal{A}(\mathcal{S}) - \mathbf{Y}\|_F^2 + \frac{\omega}{2}\left\|\mathcal{S} - \mathcal{P}_\Omega(\mathcal{T}^k) - \frac{1}{\omega}\mathcal{E}^k\right\|_F^2 \tag{16}$$

$$\mathcal{T}^{k+1} = \underset{\mathcal{T}}{\operatorname{argmin}} \rho\|\mathcal{T}\|_{\mathrm{TV}} \times \frac{\omega}{2}\left\|\mathcal{T} - \mathcal{P}_\Omega^\dagger\left(\mathcal{S}^{k+1} - \frac{1}{\omega}\mathcal{E}^k\right)\right\|_F^2 \tag{17}$$

$$\mathcal{E}^{k+1} = \mathcal{E}^k - \omega\left(\mathcal{S}^{k+1} - \mathcal{P}_\Omega(\mathcal{T}^{k+1})\right). \tag{18}$$

1) Solve $\mathcal{S}$: Equation (16) is a quadratic optimization problem with a closed solution satisfies

$$\mathcal{A}^\dagger(\mathcal{A}(\mathcal{S})) + \omega\mathcal{S} = \mathcal{E}^k + \omega\mathcal{P}_\Omega(\mathcal{T}^k) + \mathcal{A}^\dagger(\mathbf{Y}). \tag{19}$$

According to the Sherman–Morrison formula [18], the explicit expression of the solution can be written as

$$\mathcal{S}^{k+1} = \mathcal{P}_\Omega(\mathcal{T}^k) + \frac{1}{\omega}\mathcal{E}^k + \mathcal{A}^\dagger\left((\mathbf{Y} - \mathcal{A}(\mathcal{P}_\Omega(\mathcal{T}^k) + \frac{1}{\omega}\mathcal{E}^k)) \oslash (\omega + \mathcal{A}(\mathcal{C}))\right) \tag{20}$$

where $\oslash$ denotes the Hadamard (element-wise) division.

2) Solve $\mathcal{T}$: Equation (17) can be solved by computing the proximal projection operator [48]

$$\mathcal{T}^{k+1} = \mathrm{prox}_{\rho/\omega\|\cdot\|_{\mathrm{TV}}}\left(\mathcal{P}_\Omega^\dagger(\mathcal{S}^{k+1} - \frac{1}{\omega}\mathcal{E}^k)\right). \tag{21}$$

The restored smooth grayscale image can be obtained after a few iterations

$$\hat{\mathbf{Y}}_S = \frac{1}{B}\sum_b \mathcal{T}^{(b)}. \tag{22}$$

The pseudocode for solving (14) is summarized in Algorithm 1.

B. Nonlocal HSI Blocks Clustering

An HSI cube of a typical natural scene, considered as a 3-D tensor $\mathcal{X}$, is self-similar [28], thus a local block extracted from $\mathcal{X}$ can be considered similar to the surrounding blocks. This property can significantly assist in HSI reconstruction. As mentioned in Section V-A, the estimated grayscale image $\hat{\mathbf{Y}}_S$ preserves the inherent spatial structure of the HSI, and therefore, it can assist us in finding and clustering the similar HSI blocks. To be concrete, the estimated grayscale image $\hat{\mathbf{Y}}_S$ is initially partitioned into N overlapping patches with a spatial size of $d \times d$ and an overlap size of l. Each patch can be represented by a submatrix $\mathbf{Z}_i \in \mathbb{R}^{d\times d}, i = 1, 2, \ldots, N$. To maintain consistent pixel coverage during the extraction of image blocks, d is selected as a n times of $d-l$, where n is an integer. Subsequently, from the $L \times L$ surrounding patches of $\mathbf{Z}_i$, M_i patches with the minimum Euclidean distances from $\mathbf{Z}_i$ are found, forming a cluster $\mathcal{W}_i \in \mathbb{R}^{d\times d\times M_i}$ composed of these patches. Ultimately, a set of clusters, conceptualized as 4-D tensors $\{\mathcal{B}_i \in \mathbb{R}^{d\times d\times B\times M_i}\}_{i=1}^N$, is generated from the HSI $\mathcal{X}$, and is associate with $\{\mathcal{W}_i\}_{i=1}^N$ based on spatial location, where $\mathcal{B}_i$ is consisted of 3-D tensors $\{\mathcal{D}_i \in \mathbb{R}^{d\times d\times B}\}_{i=1}^{M_i}$. It's worth noting that HSI blocks within the same cluster exhibit analogous spatial–spectral structures, and thus each 4-D tensor is of *low rank*.

To apply the self-similarity prior to HSIs reconstruction, a low-rank regularizer of $\mathcal{B}_i$ is introduced. The optimization model of HSI reconstruction can be formulated as

$$\min_{\mathcal{X}} \sum_{i=1}^N \|\mathcal{B}_i - \mathcal{Z}_i\|_F + \frac{\beta}{2}\|\mathbf{Y} - \mathcal{A}(\mathcal{P}_\Omega(\mathcal{X}))\|_F^2 \quad \text{s.t. } \mathrm{rank}(\mathcal{Z}_i) = r, \quad i = 1, 2, \ldots, N \tag{23}$$

where β is a penalty parameter. As mentioned above, the HSI block cluster $\mathcal{B}_i$ is constructed from $\mathcal{X}$, and exhibits strong low-rank characteristics, $\mathcal{Z}_i$ is a rank-r tensor of the same size as $\mathcal{B}_i$.

C. Optimization Procedure

An auxiliary variable $\mathcal{V} \in \mathbb{R}^{(m+\alpha(B-1))\times n\times B}$ is introduced in (23) to adopt the ADMM for solving the aforementioned problem. The problem model is rewritten in the following equivalently expression:

$$\min_{\mathcal{X},\mathcal{V}} \sum_{i=1}^N \|\mathcal{B}_i - \mathcal{Z}_i\|_F + \frac{\beta}{2}\|\mathbf{Y} - \mathcal{A}(\mathcal{V})\|_F^2 \quad \text{s.t. } \mathcal{V} = \mathcal{P}_\Omega(\mathcal{X}) \ \mathrm{rank}(\mathcal{Z}_i) = r, \quad i = 1, 2, \ldots, N. \tag{24}$$

Therefore, the objective function can be expressed in the form of an augmented Lagrangian function

$$\min_{\mathcal{X},\mathcal{V},\mathcal{M}} \mathcal{L}(\mathcal{X}, \mathcal{V}, \mathcal{M}) := \sum_{i=1}^N \|\mathcal{B}_i - \mathcal{Z}_i\|_F + \frac{\beta}{2}\|\mathbf{Y} - \mathcal{A}(\mathcal{V})\|_F^2 - \langle\mathcal{M}, \mathcal{V} - \mathcal{P}_\Omega(\mathcal{X})\rangle + \frac{\gamma}{2}\|\mathcal{V} - \mathcal{P}_\Omega(\mathcal{X})\|_F^2 \tag{25}$$

where $\mathcal{M} \in \mathbb{R}^{(m+\alpha(B-1))\times n\times B}$ is a Lagrangian multiplier, γ is the penalty parameter. For simplicity, the rank constraint on $\mathcal{Z}_i$ is omitted. Then, within the framework of ADMM methodology, the optimization subproblem can be solved as follows.

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1) Solve $\mathcal{V}$: The subproblem of $\mathcal{V}$ can be written as

$$\mathcal{V}^{k+1} = \underset{\mathcal{V}}{\operatorname{argmin}} \frac{\beta}{2}\|\mathbf{Y} - \mathcal{A}(\mathcal{V})\|_F^2 + \frac{\gamma}{2}\left\|\mathcal{V} - \mathcal{P}_\Omega(\mathcal{X}^k) - \frac{1}{\gamma}\mathcal{M}^k\right\|_F^2. \tag{26}$$

Equation (26) has a closed-form solution satisfies

$$\gamma\mathcal{V} + \beta\mathcal{A}^\dagger(\mathcal{A}(\mathcal{V})) = \mathcal{M}^k + \gamma\mathcal{P}_\Omega(\mathcal{X}^k) + \beta\mathcal{A}^\dagger(\mathbf{Y}). \tag{27}$$

Similar to the solution process of (16), the result can be written as

$$\mathcal{V}^{k+1} = \mathcal{P}_\Omega(\mathcal{X}^k) + \frac{1}{\gamma}\mathcal{M}^k + \mathcal{A}^\dagger\left(\left(\mathbf{Y} - \mathcal{A}\left(\mathcal{P}_\Omega(\mathcal{X}^k) + \frac{1}{\gamma}\mathcal{M}^k\right)\right) \oslash \left(\frac{\gamma}{\beta} + \mathcal{A}(\mathcal{C})\right)\right). \tag{28}$$

2) Solve $\mathcal{X}$: The optimal solution of $\mathcal{X}$ is given through solving the following subproblem:

$$\mathcal{X}^{k+1} = \underset{\mathcal{X}}{\operatorname{argmin}} \sum_{i=1}^{N}\|\mathcal{B}_i - \mathcal{Z}_i\|_F + \frac{\gamma}{2}\left\|\mathcal{P}_\Omega^\dagger\left(\mathcal{V}^{k+1} - \frac{1}{\gamma}\mathcal{M}^k\right) - \mathcal{X}\right\|_F^2 \tag{29}$$

where again $\{\mathcal{B}_i\}_{i=1}^N$ is the set of block clusters extracted from $\mathcal{X}$, and it can be written as

$$\{\mathcal{B}_i\}_{i=1}^N = \text{Extract}(\mathcal{X}). \tag{30}$$

Equation (30) can be converted to the following problem:

$$\mathcal{X}^{k+1} = \underset{\mathcal{X}}{\operatorname{argmin}} \sum_{i=1}^{N}\|\mathcal{B}_i - \mathcal{Z}_i\|_F + \frac{\gamma}{2}\left\|\mathcal{Q}^k - \mathcal{X}\right\|_F^2 \tag{31}$$

where $\mathcal{Q}^k = \mathcal{P}_\Omega^\dagger(\mathcal{V}^{k+1} - (1/\gamma)\mathcal{M}^k)$. As described in Section V-B, the patch size d is n times of $d - l$. Since the number of HSI blocks is much larger than the number of edge blocks, $N \gg n$, the following equation is approximately established:

$$\left\|\mathcal{Q}^k - \mathcal{X}\right\|_F^2 \approx \frac{1}{n}\sum_{i=1}^{N}\left\|\mathcal{O}_i^k - \mathcal{D}_i\right\|_F^2 \tag{32}$$

where $\mathcal{O}_i^k$ denotes the ith block of $\mathcal{Q}_i^k$ associated with $\mathcal{D}_i$ based on spatial location. Subsequently, (31) can be relaxed as

$$\begin{aligned}\mathcal{B}_i^{k+1} &= \underset{\mathcal{B}_i}{\operatorname{argmin}}\ \|\mathcal{B}_i - \mathcal{Z}_i\|_F + \frac{\gamma}{2n}\sum_{i=1}^{M_i}\left\|\mathcal{O}_i^k - \mathcal{D}_i\right\|_F^2 \\ &= \underset{\mathcal{B}_i}{\operatorname{argmin}}\ \|\mathcal{B}_i - \mathcal{Z}_i\|_F + \frac{\gamma}{2n}\left\|\mathcal{J}_i^k - \mathcal{B}_i\right\|_F^2\end{aligned} \tag{33}$$

where $\{\mathcal{J}_i^k\}_{i=1}^N = \text{Extract}(\mathcal{Q}^k)$. The solution of (33) can be expressed as

$$\mathcal{X}^{k+1} = \text{Joint}\left(\left\{\mathcal{F}_{\text{CP}}(\mathcal{J}_i^k, r)\right\}_{i=1}^N\right) \tag{34}$$

where Joint$(\cdot)$ denotes the inverse operation of Extract$(\cdot)$. $\mathcal{F}_{\text{CP}}(\mathcal{J}, r)$ represents the CP-rank r approximation of $\mathcal{J}$ and it can be calculated by alternating least squares method [48].

3) Updating Multipliers: In accordance with the ADMM method, the multipliers are updated using

$$\mathcal{M}^{k+1} = \mathcal{M}^k - \gamma(\mathcal{V}^{k+1} - \mathcal{P}_\Omega(\mathcal{X}^{k+1})). \tag{35}$$

Summarizing the above description, the proposed algorithm, NLRT, is summarized in Algorithm 2 as follows.

Algorithm 2 NLRT Algorithm for HSI Reconstruction

Input: tensor $\mathcal{C}$, matrix $\mathbf{Y}$, trade-off parameter $\beta > 0$, penalty parameter $\gamma > 0$, CP-rank r, patch size d, overlap size l.
Initialize: $\mathcal{X}^1, \mathcal{V}^1, \mathcal{M}^1$.

```
Estimate Ŷ_S by Algorithm 1;
Divide measurement Ŷ_S into N patches;
Group similar patches into clusters {W_i}_{i=1}^N;
for k:=1 to Max_iters do
    Update V^{k+1} using Eq. (28);
    Calculate Q^k using Eq. (31);
    Extract blocks of Q^k;
    Group blocks into clusters {J_i^k}_{i=1}^N based on {W_i}_{i=1}^N;
    for i = 1 to N do
        B_i^{k+1} = F_CP(J_i^k, r);
    end for
    Joint block clusters {B_i^{k+1}}_{i=1}^N and update X^{k+1} using Eq. (34);
    Update M^{k+1} using Eq. (35);
end for
```

Output: The recovered hyperspectral image $\mathcal{X}$.

VI. Numerical Experiment

The effectiveness of the proposed NLRT method for HSIs reconstruction in the SD-CASSI system was evaluated in this section. The evaluation included numerical experiments using both synthetic HSI datasets and real HSI datasets. In addition, we conducted a comparison assessment with SOTA methods and analyzed the convergence of the proposed NLRT method. All experiments are run on a platform with Intel Xeon w7-2495X and NVIDIA RTX A6000.

A. Experimental Datasets

1) Synthetic Datasets: To illustrate the adaptability of the proposed method, four hyperspectral datasets were employed to perform simulated experiments in this article. The KAIST[1] [49] dataset consisted of 30 HSI scenes, each with a spatial resolution of 2704 × 3376 pixels and 31 spectral channels from 420 to 720 nm. The ICVL[2] [50] dataset comprised 200 HSI scenes, featuring a spatial extent of 1392 × 1300 pixels and 31 spectral channels from 400 to 700 nm. The CAVE[3] [51] dataset comprises 32 hyperspectral scenes across 31 wavelength bands, spanning from 400 to 700 nm with a step of 10 nm. Each band is characterized by a pixel dimension of 512 × 512. The Harvard[4] [52] dataset comprises

[1] https://vclab.kaist.ac.kr/siggraphasia2017p1/kaistdataset.html
[2] https://icvl.cs.bgu.ac.il/hyperspectral
[3] https://www1.cs.columbia.edu/CAVE/databases/multispectral
[4] http://vision.seas.harvard.edu/hyperspec

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TABLE I

COMPARISONS BETWEEN NLRT METHOD AND DL-BASED METHODS ON TEN SCENES (S1–S10). THE AVERAGE PSNR, SSIM, AND SAM ARE LISTED

Methods		S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	avg
GAP-Net	PSNR	33.29	34.33	41.41	31.46	33.53	33.05	18.00	26.33	29.25	33.12	30.66
	SSIM	0.9010	0.9324	0.9725	0.9214	0.9201	0.8903	0.6734	0.8322	0.8141	0.8788	0.8671
	SAM	13.07	8.60	9.55	7.87	8.76	12.67	23.94	17.06	15.05	15.33	13.33
ADMM-Net	PSNR	33.64	35.10	41.51	31.85	33.77	33.78	19.95	28.26	30.37	34.79	31.60
	SSIM	0.9029	0.9343	0.9721	0.9241	0.9209	0.9021	0.7502	0.8667	0.8249	0.9066	0.8862
	SAM	13.05	8.13	9.37	7.90	8.29	11.70	17.45	12.78	12.96	12.42	11.43
HDNet	PSNR	35.73	36.14	42.78	32.73	34.93	34.19	20.08	27.67	29.87	34.37	32.22
	SSIM	0.9424	0.9478	0.9776	0.9483	0.9471	0.8810	0.7598	0.8546	0.8261	0.8781	0.8928
	SAM	8.16	5.65	5.82	5.07	6.35	14.31	22.00	16.32	16.22	16.28	11.65
MST-L	PSNR	36.20	36.48	42.36	33.00	35.12	34.85	20.02	28.52	31.19	35.64	32.59
	SSIM	0.9479	0.9538	0.9771	0.9486	0.9462	0.9166	0.7830	0.8820	0.8594	0.9230	0.9071
	SAM	8.13	6.09	7.43	5.80	7.47	9.59	19.02	12.03	11.64	10.70	10.01
CST-L	PSNR	36.55	37.55	42.67	33.41	36.03	34.72	20.28	27.70	30.44	35.35	32.80
	SSIM	0.9541	0.9628	0.9789	0.9547	0.9561	0.9139	0.7606	0.8752	0.8567	0.9121	0.9067
	SAM	6.76	4.71	5.75	4.46	5.73	11.19	20.35	13.88	12.78	12.34	9.97
DAUHST-9stg	PSNR	**39.15**	41.27	47.01	**35.88**	40.14	38.07	23.62	31.69	33.24	39.00	36.25
	SSIM	**0.9688**	**0.9748**	0.9886	**0.9709**	0.9738	0.9569	0.8190	0.9205	0.8890	0.9572	0.9383
	SAM	6.04	3.45	5.57	3.62	4.37	5.46	11.21	7.67	8.24	6.27	6.22
BIRNAT	PSNR	37.98	40.84	47.77	35.50	39.56	38.76	28.17	35.25	34.68	39.40	37.31
	SSIM	0.9597	0.9745	0.9899	0.9658	0.9734	0.9622	0.8916	0.9351	**0.8935**	0.9545	0.9488
	SAM	6.79	3.32	3.86	3.93	**3.93**	4.81	4.44	5.15	6.44	5.14	4.53
NLRT(ours)	PSNR	37.50	**41.42**	**48.60**	34.34	**42.23**	**40.46**	**33.22**	**37.13**	**35.35**	**42.37**	**38.92**
	SSIM	0.9546	0.9746	**0.9910**	0.9486	**0.9785**	**0.9681**	**0.9419**	**0.9362**	0.8826	**0.9646**	**0.9565**
	SAM	**5.80**	**2.63**	**3.19**	**3.49**	4.06	**3.12**	**1.40**	**3.26**	**4.66**	**2.46**	**3.13**

Best results are red in **bold**, *second best results are underline.*



Fig. 5. Ten testing scenes for numerical experimental with spatial size of 256 × 256. The first row from left to right are scenes 1–5, and the second row are scenes 6–10.

50 hyperspectral scenes capturing both indoor and outdoor environments. These scenes are characterized by 31 spectral bands ranging from 420 to 720 nm at intervals of 10 nm. The spatial dimensions of this dataset measure 1040 × 1392 pixels.

Specifically, comparisons were drawn with DL-based methods on the KAIST and ICVL datasets, while optimization-based and PnP-based methods were contrasted on the CAVE and Harvard datasets. Similar to TSA-Net[5] [53], five scenes were selected from KAIST with a spatial size of 256 × 256. In addition, another five scenes, with 28 spectral channels ranging from 400 to 670 nm, were cropped from ICVL with a spatial size of 256 × 256. The ten scenes, used for testing the performance of algorithms, are shown in Fig. 5.

2) Real Datasets: Five real HSIs collected by the SD-CASSI system developed in [53] are used for testing. The measurements were captured with a spatial resolution of 660 × 714. The code pattern employed was a matrix with a spatial resolution of 660 × 660 and contains noise.

[5]https://github.com/mengziyi64/TSA-Net

B. Implementation Details

When simulating the process of HSIs compression, we use a random coded pattern with a transmittance of 50%, and the dispersive step $\alpha = 2$. For our methods, we set the regularization parameters $\rho = 5e-6$, the tradeoff parameter $\beta = 1e-3$, the CP-rank $r = 9$, the penalty parameters $\gamma = 1e-3$ and $\omega = 1e-5$, respectively. For the simulated dataset, we set the patch size $d = 10$ with the overlapping size $l = 5$, and for the edge pixels of the image, we additionally extract image blocks so that the extracted image blocks contain all pixels of the image. The stopping criterion for the algorithm is set to reach a maximum of 300 iterations.

C. Performance Evaluation Measures

Peak signal-to-noise ratio (PSNR), structural similarity (SSIM), spectral angle mapper (SAM), and computation time (time) are chosen as evaluation metrics to evaluate the algorithm's performance. To facilitate comparisons, we calculate the average PSNR (dB) and SSIM values across all spectral bands, as well as the average SAM (°) across all spatial points. A higher PSNR indicates higher reconstruction accuracy, while a higher SSIM corresponds to better spatial reconstruction quality. Conversely, a smaller SAM value denotes the spectral reconstruction quality, and shorter time (s) indicate higher reconstruction efficiency.

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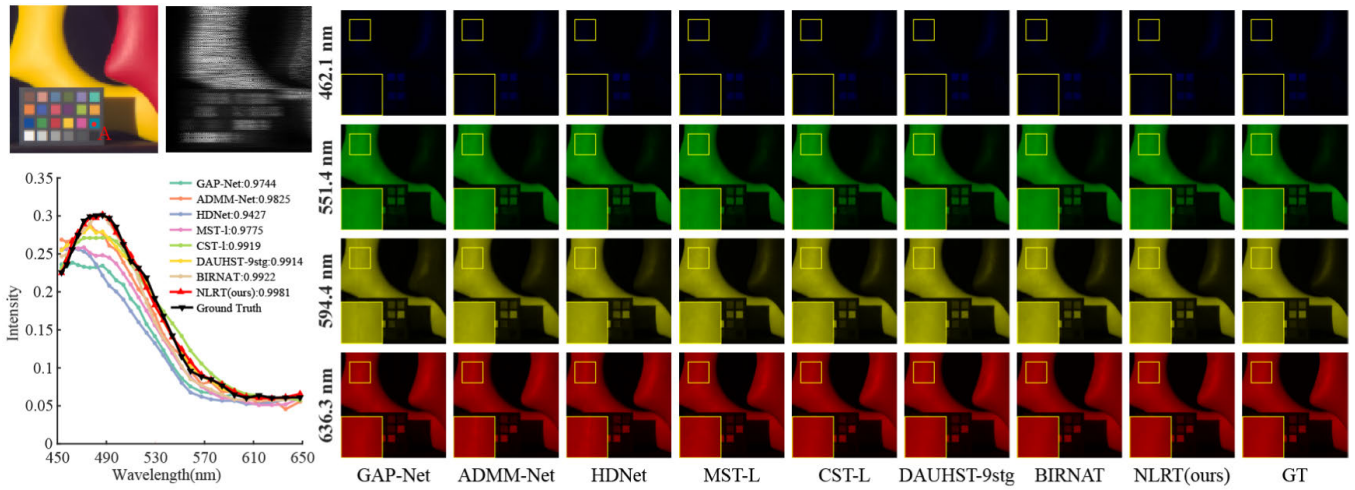


Fig. 6. Visual presentation of the reconstruction results for scene 2. (Left) Reconstructed spectral curve at point A in the RGB image, along with the correlation coefficient with the GT. (Right) Reconstruction results for four classic spectral bands, comparing eight reconstruction methods with the GT.

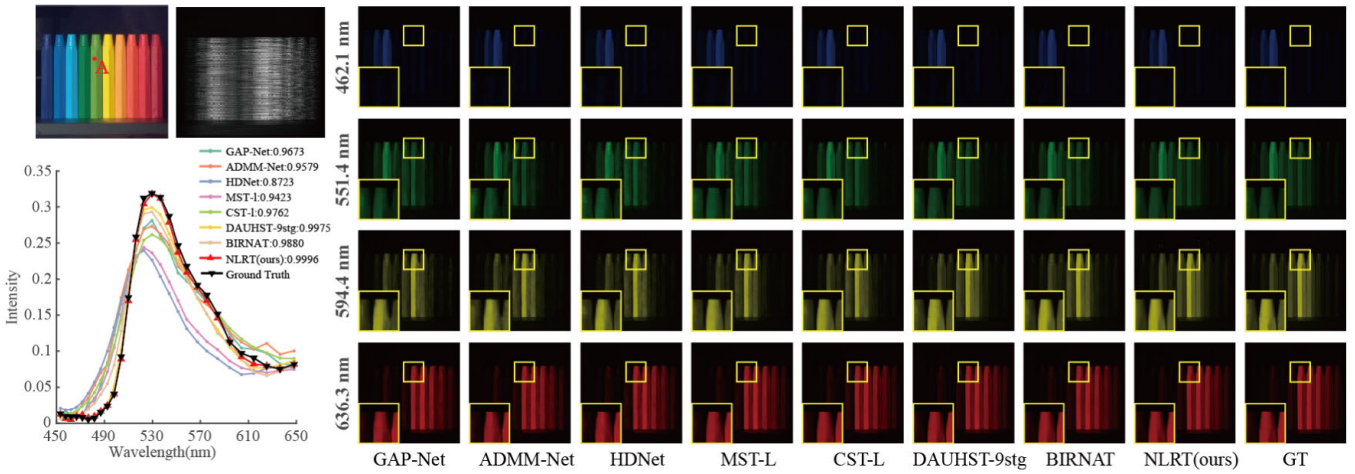


Fig. 7. Visual presentation of the reconstruction results for scene 5. (Left) Reconstructed spectral curve at point A in the RGB image, along with the correlation coefficient with the GT. (Right) Reconstruction results for four classic spectral bands, comparing eight reconstruction methods with the GT.

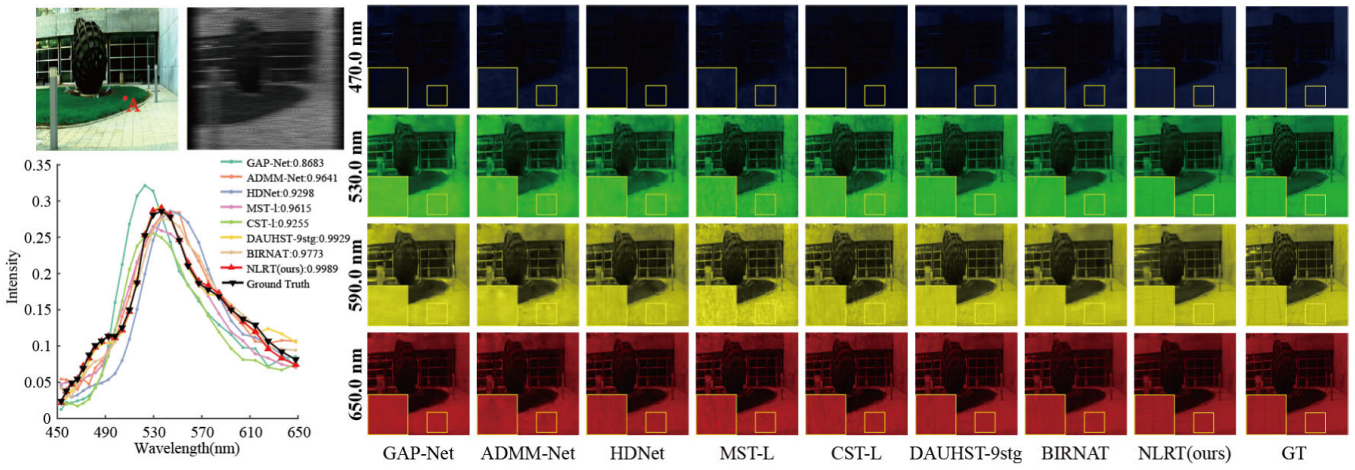


Fig. 8. Visual presentation of the reconstruction results for scene 10. (Left) Reconstructed spectral curve at point A in the RGB image, along with the correlation coefficient with the GT. (Right) Reconstruction results for four classic spectral bands, comparing eight reconstruction methods with the GT.

D. Synthetic Data Results

1) Compared With DL-Based Methods: To illustrate the performance of the proposed NLRT method, we implemented seven methods for comparison, including four fully learned methods (HDNet [54], MST[6] [21], CST [20], and

[6]https://github.com/caiyuanhao1998/MST

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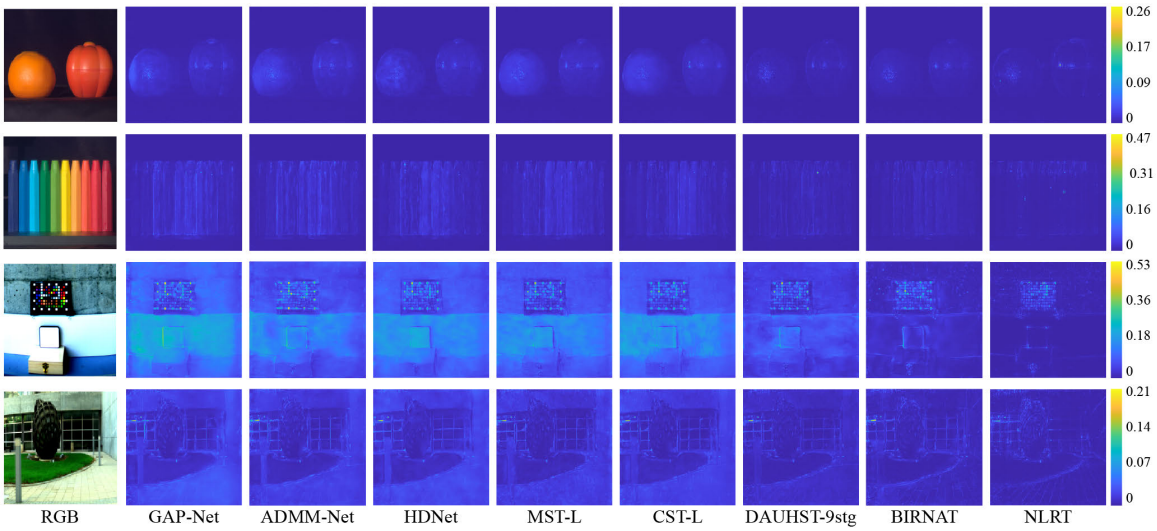


Fig. 9. Visual analysis of the error map in reconstructed images, from top to bottom, corresponds to scenes 3, 5, 7, and 10.

TABLE II

COMPARISONS BETWEEN NLRT METHOD AND OPTIMIZATION-BASED METHODS ON TEN SCENES. THE AVERAGE PSNR, SSIM, AND SAM ARE LISTED

Methods		Face	Toy	Clay	Egyptian	Cloth	Imgh0	Img1	Imgd3	Imgc4	Imgb8	avg
GAP-TV	PSNR	34.06	25.99	35.24	34.35	26.11	32.95	28.04	29.82	27.94	26.15	30.0652
	SSIM	0.9229	0.8112	0.9105	0.9093	0.5696	0.7987	0.7929	0.8267	0.7257	0.6176	0.7885
	SAM	8.63	10.57	12.37	15.64	9.30	4.54	4.90	7.72	6.58	7.26	0.1527
	time	479	712	714	463	719	500	502	496	493	466	554
DeSCI	PSNR	36.30	26.71	39.25	35.96	26.07	33.25	29.18	30.85	29.14	26.44	31.3148
	SSIM	0.9495	0.8509	0.9689	0.9441	0.5792	0.8100	0.8294	0.8747	0.7853	0.6493	0.8241
	SAM	7.13	9.01	7.94	11.18	9.43	4.76	4.52	6.85	6.09	7.21	7.41
	time	3450	4850	4862	3470	4820	3448	3430	2463	3415	2432	3664
TV-FFDNet	PSNR	33.92	25.40	36.66	34.86	22.17	34.28	28.36	30.68	28.69	26.91	30.19
	SSIM	0.9407	0.8327	0.9568	0.9318	0.5185	0.8462	0.8042	0.8575	0.7677	0.6651	0.8121
	SAM	7.12	8.77	8.85	14.91	8.70	3.90	4.53	6.25	5.54	6.26	7.48
	time	**273**	**270**	**271**	**278**	**273**	**272**	**263**	**264**	**258**	**259**	**268**
PnP-DIP	PSNR	34.62	28.13	37.20	35.27	24.12	33.55	30.82	33.51	30.56	27.49	31.53
	SSIM	0.9113	0.8053	0.8969	0.9053	0.5995	0.8171	0.8205	0.8763	0.7994	0.6928	0.8124
	SAM	12.40	17.09	15.02	22.60	10.14	7.28	4.82	8.98	6.84	7.72	11.29
	time	28208	14842	20132	10673	19049	11163	11217	9483	10739	11747	13025
NLRT(ours)	PSNR	**40.75**	**31.43**	**45.71**	**39.09**	**26.66**	**40.86**	**33.92**	**36.95**	**34.55**	**29.45**	**35.94**
	SSIM	**0.9732**	**0.9258**	**0.9848**	**0.9669**	**0.6088**	**0.9551**	**0.9085**	**0.9430**	**0.9040**	**0.7654**	**0.8936**
	SAM	**5.21**	**5.70**	**6.44**	**10.44**	**8.35**	**2.17**	**2.16**	**3.47**	**2.64**	**4.20**	**5.08**
	time	7396	7683	7445	7536	5765	4657	4467	5128	4802	5434	6031

Best results are red in **bold**, *second best results are underline*.

BIRNAT [22]) and three model-based methods (GAP-Net [23], ADMM-Net [24], and DAUHST [25]). For fairness, we tested all methods with the same parameter settings, including encoding transmittance and dispersion step.

Table I presents the performance of our method compared to seven DL-based methods across the aforementioned ten synthetic scenes. It can be seen that our method demonstrates remarkable results, achieving an average PSNR of 38.92 dB, an average SSIM of 0.9565, and an average SAM of 3.13°. In particular, NLRT outperforms two recent SOTA approaches, DAUHST-9stg and BIRNAT, by improving PSNR by 2.67 and 1.61 dB in terms of PSNR, thereby confirming the effectiveness of our method. Furthermore, the proposed NLRT method exhibits a significant improvement in SAM, implying that the reconstruction results of NLRT possess better spectral fidelity.

In the context of scenes derived from the KAIST dataset (scenes 1–5), our method exhibits comparable performance to that of DAUHST and BIRNAT. Specifically, for scenes 2, 3, and 5, our method outperforms the existing SOTA methods. Although our proposed method falls slightly behind DAUHST and BIRNAT in terms of PSNR and SSIM, we still maintain a lead over them in SAM. Furthermore, our approach retains a substantial advantage over earlier DL-based methods. In scenes selected from the ICVL dataset, our method significantly outperforms the SOTA methods. The only exception is scene 9, where our SSIM is 0.0109 lower than that of BIRNAT, but it still remains at a similar level. It is worth noting that our method consistently maintains a lead in

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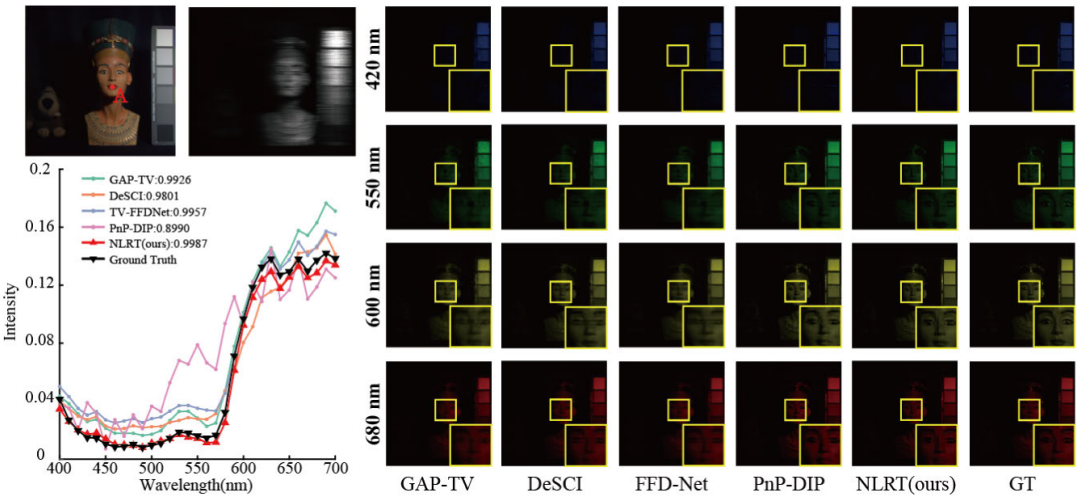


Fig. 10. Visual presentation of the reconstruction results for CAVE-Egyptian. (Left) Reconstructed spectral curve at point A in the RGB image, along with the correlation coefficient with the GT. (Right) Reconstruction results for four classic spectral bands, comparing five reconstruction methods with the GT.

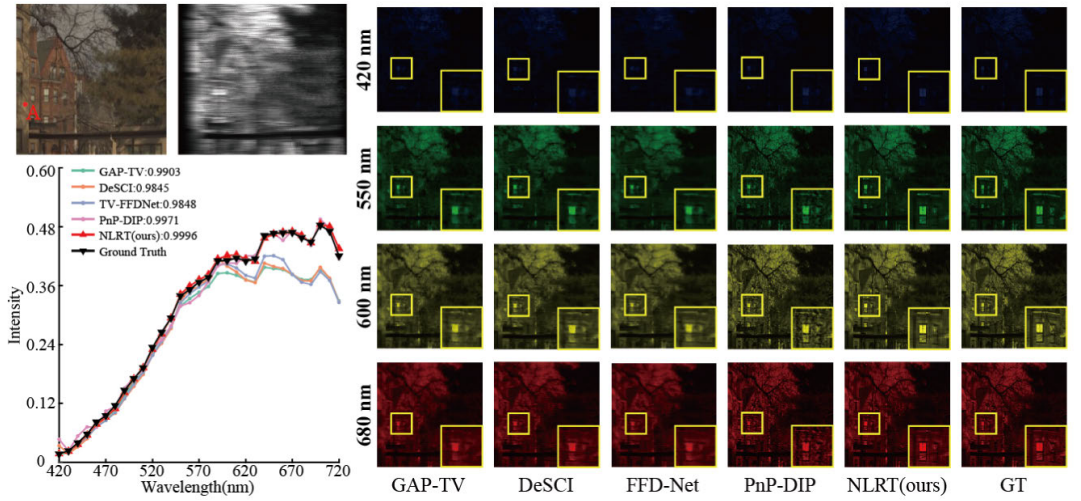


Fig. 11. Visual presentation of the reconstruction results for Harvard-imgh0. (Left) Reconstructed spectral curve at point A in the RGB image, along with the correlation coefficient with the GT. (Right) Reconstruction results for four classic spectral bands, comparing five reconstruction methods with the GT.

SAM, indicating its capability to precisely reconstruct spectral information. For HSIs, accurate spectral information is of paramount importance.

The reconstruction results of all methods on scenes 2, 5, and 10 are shown in Figs. 6–8, respectively. We selected four spectral bands from the HSI and transformed them into RGB representations using a wavelength-to-RGB conversion technique to facilitate visualization of the spectral reconstruction outcomes. Moreover, we focused on particular regions of interest (ROIs) outlined by yellow rectangles, providing enlarged views of these areas. Compared to the other methods, Our method exhibits the ability to reconstruct finer details. Within each scene, we pinpointed a critical reference point and generated spectral curves to facilitate comparative analysis. The level of correlation with respect to the ground truth is indicated in the legend, serving as a quantitative measure of spectral fidelity. It can be observed that our method provides more precise results than the other methods.

As shown in Fig. 9, we illustrate the error distributions for scenes 3, 5, 7, and 10. It is evident that our method achieves the best reconstruction performance for spatial regions with similar spectral distributions, indicating that the NLRT holds a powerful capability to effectively leverage self-similarity information of the natural images.

2) Compared With Optimization-Based Methods: To quantitatively compare the proposed NLRT method with optimization-based methods, we have chosen four optimization-based approaches, which include two methods based on handcrafted priors, GAP-TV [17] and DeSCI[7] [18], as well as two methods based on deep priors, TV-FFDNet[8] [42] and PnP-DIP[9] [41]. Table II presents the performance of our method as well as those of four optimization-based methods across ten synthetic scenes. The results indicate that although our proposed NLRT method may not excel in reconstruction time, it significantly improves the reconstruction quality of HSI.

[7]https://github.com/XiaoYangLiu-FinRL/DeSCI
[8]https://github.com/ucker/SCI-TV-FFDNet
[9]https://github.com/mengziyi64/CASSI-Self-Supervised

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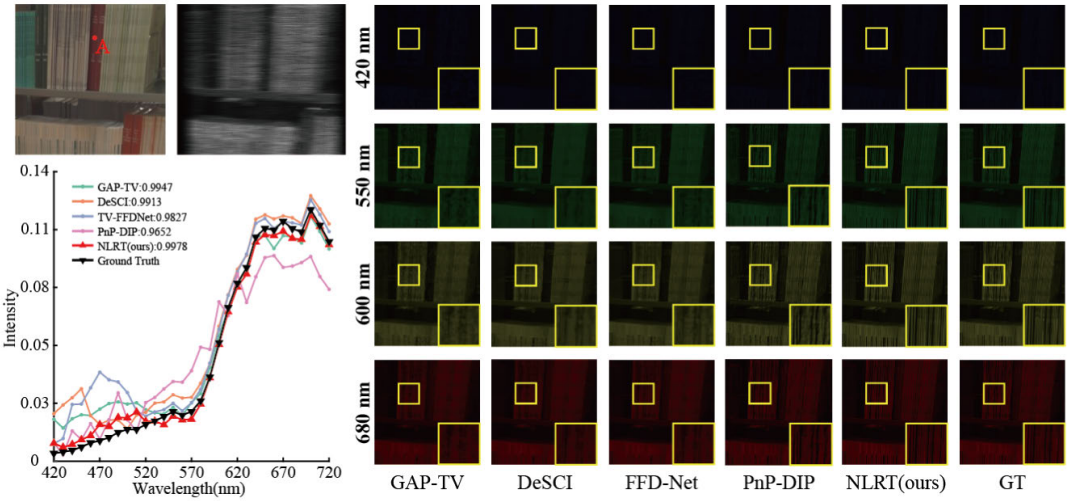


Fig. 12. Visual presentation of the reconstruction results for Harvard-imgb8. (Left) Reconstructed spectral curve at point A in the RGB image, along with the correlation coefficient with the GT. (Right) Reconstruction results for four classic spectral bands, comparing five reconstruction methods with the GT.

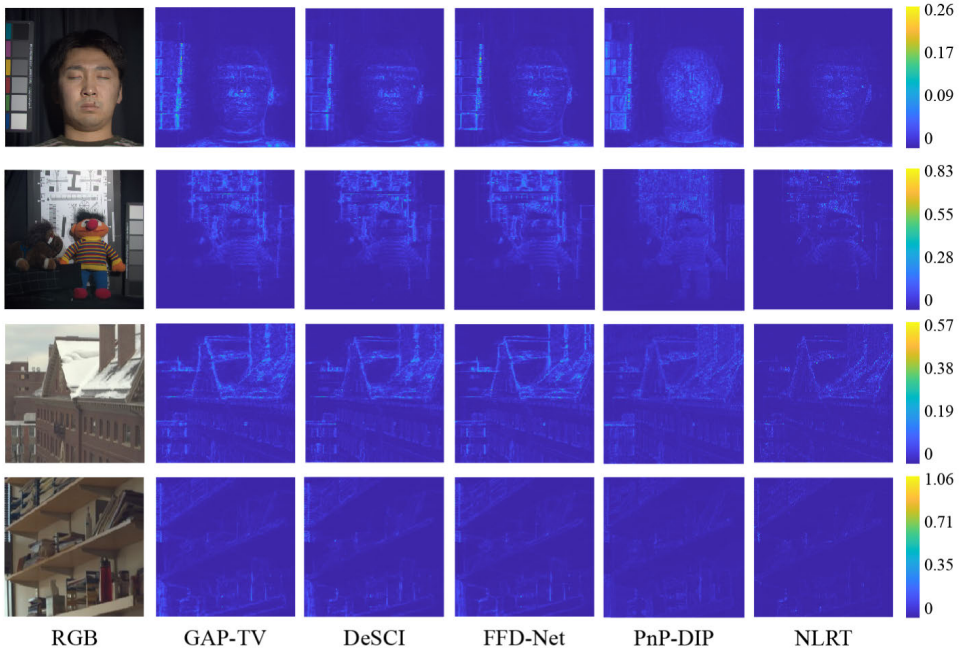


Fig. 13. Visual analysis of the error map in reconstructed images, from top to bottom, corresponds to CAVE-Face, CAVE-Toy, Harvard-img1, and Harvard-imgd3.

The visual results of the reconstruction of three scenes—CAVE-Egyptian, Harvard-imgh0, and Harvard-imgb8 are shown in Figs. 10–12, respectively, depict the visual results of the reconstruction of three scenes: CAVE-Egyptian, Harvard-imgh0, and Harvard-imgb8. In terms of spatial details, the NLRT method retrieves more intricate features. At selected points in these scenes, the spectral curves reconstructed by NLRT exhibit the highest agreement with ground truth spectra.

Fig. 13 illustrates the distribution of mean square errors between the proposed NLRT method and optimization-based methods across four scenes. It is evident that greater errors mostly occur at object edges. Nonetheless, the NLRT method achieves high-precision HSI reconstruction within the objects.

E. Real Data Results

We applied the NLRT method directly to the real dataset using the same parameter settings. Fig. 14 shows the results of real HSIs data for 3 out of 28 bands (551.4, 594.4, and 636.3 nm) using the NLRT method, along with results from five other methods. It can be observed that our method retains more details compared to the DAUHST method, demonstrating the effectiveness of our approach on real datasets.

F. Convergence Analysis

We provide an empirical convergence analysis on the experimental dataset, using the relative change relChgX := $(\|\boldsymbol{\mathcal{X}}^k - \boldsymbol{\mathcal{X}}^{k-1}\|_F^2 / \|\boldsymbol{\mathcal{X}}^{k-1}\|_F^2)$ and the relative err relErrX := $(\|\boldsymbol{\mathcal{X}}^k - \boldsymbol{\mathcal{X}}\|_F^2 / \|\boldsymbol{\mathcal{X}}\|_F^2)$ for quantitative evaluation, respectively.

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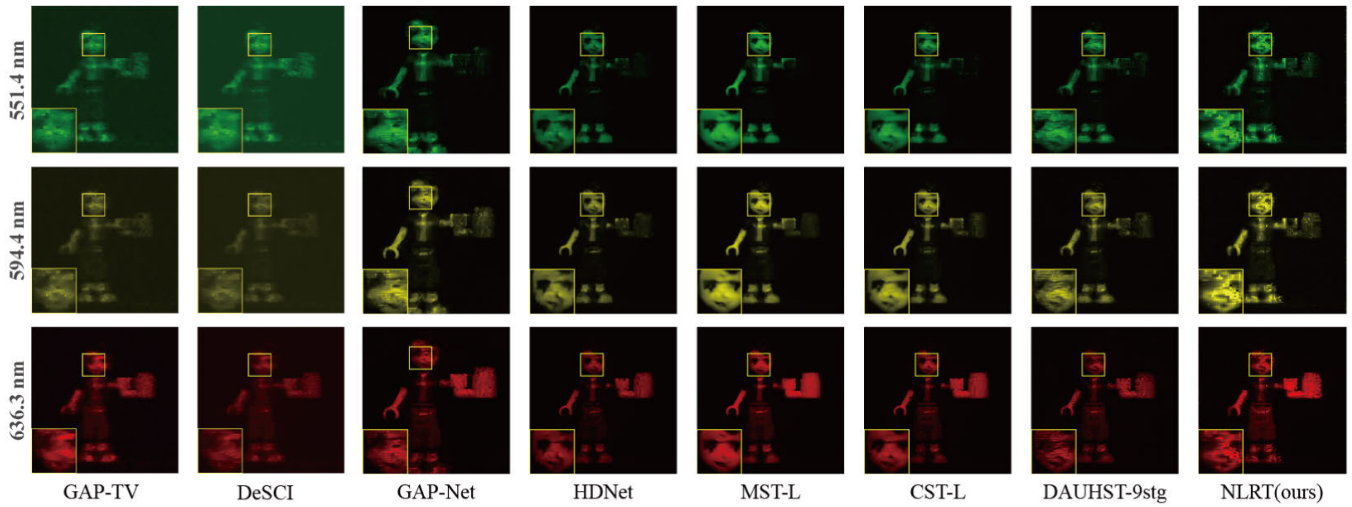


Fig. 14. Reconstructed spectral images of real dataset. We present the visualization of three spectral bands (with wavelengths at 551.4, 594.4, and 636.3 nm) using six reconstruction methods.

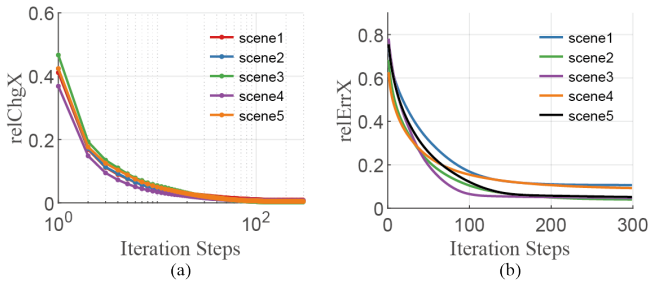


Fig. 15. Empirical analysis of algorithm convergence of scenes 1–5. (a) relChgX and (b) relErrX.

In Fig. 15, we present the convergence analysis for five scenes. The results show that the proposed NLRT method can achieve convergence within 300 iterations and exhibits good stability. Therefore, the algorithm's convergence can be assessed by setting the relChgX threshold.

G. Discussion

It can be observed that DL-based methods perform less effectively on the ICVL dataset compared to their performance on the KAIST dataset. One possible reason for this discrepancy is that these DL-based methods may not have been optimized for real-world natural datasets like ICVL. Compared to DAUHST, the BIRNAT method continues to exhibit excellent results on the ICVL dataset. One possible reason for this is that, similar to our proposed NLRT method, the BIRNAT method takes into account the spectral shearing characteristics in the SD-CASSI system.

Existing optimization-based approaches overlooked the inherent shearing degradation issue in SD-CASSI systems. Simply transplanting the reconstruction algorithm of the DD-CASSI system into the SD-CASSI system leads to inferior reconstruction performance. The proposed NLRT method introduces shearing degradation constraints within the two-step projection model, further refining the feasible domain range and enhancing algorithmic precision.

VII. Conclusion

In this article, we proposed a tensor-based two-step projection imaging model of SD-CASSI that combines the spectral shearing projection and encoding-compression projection. In contrast to the previous model, this model comprehensively depicts the transformation from the HSI of targets to the SD-CASSI measurement. As a result, in this model, the regularization term can be directly applied to the original HSI. Then, we developed the optimization framework for this NLRT method, which takes full advantage of the spatial self-similar information restored from measurement images. In addition, a CP-norm is introduced to constrain the low-rank property of the reconstructed HSI. It is noteworthy that the method NLRT is the first optimization-based method specifically developed for SD-CASSI.

Experimental results show that our NLRT method outperforms SOTA HSI reconstruction methods, which are all based on DL networks. Compared with the latest SOTA methods, our proposed NLRT method exhibits a significant lead in the SAM evaluation. Finer image partitioning can potentially lead to better results, but it also implies longer computation times.

References

[1] M. Borengasser, W. S. Hungate, and R. Watkins, *Hyperspectral Remote Sensing: Principles and Applications*. Boca Raton, FL, USA: CRC Press, 2007.

[2] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[3] G. Lu and B. Fei, "Medical hyperspectral imaging: A review," *J. Biomed. Opt.*, vol. 19, no. 1, Jan. 2014, Art. no. 010901.

[4] J. Yang, Y.-Q. Zhao, J. C. Chan, and S. G. Kong, "Coupled sparse denoising and unmixing with low-rank constraint for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1818–1833, Mar. 2016.

[5] Z. Pan, G. Healey, M. Prasad, and B. Tromberg, "Face recognition in hyperspectral images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 25, no. 12, pp. 1552–1560, Dec. 2003.

[6] A. Liu, Y. Yuan, L. Su, X. Meng, H. Shao, and Y. Jiang, "Hybrid non-sequential modeling of an image mapping spectrometer," *Appl. Opt.*, vol. 61, no. 17, pp. 5260–5268, 2022.

[7] L. Su, Y. Liu, and Y. Yuan, "Spectrum reconstruction of the light-field multimodal imager," *IEEE Access*, vol. 7, pp. 9688–9696, 2019.

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[8] C. Zhu, H. Zheng, Y. Yuan, and L. Su, "Spectrum reconstruction of multispectral light field imager based on adaptive sparse representation," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–12, 2024.

[9] M. E. Gehm, R. John, D. J. Brady, R. M. Willett, and T. J. Schulz, "Single-shot compressive spectral imaging with a dual-disperser architecture," *Opt. Exp.*, vol. 15, no. 21, pp. 14013–14027, Oct. 2007.

[10] A. A. Wagadarikar, N. P. Pitsianis, X. Sun, and D. J. Brady, "Video rate spectral imaging using a coded aperture snapshot spectral imager," *Opt. Exp.*, vol. 17, no. 8, pp. 6368–6388, 2009.

[11] X. Lin, Y. Liu, J. Wu, and Q. Dai, "Spatial–spectral encoded compressive hyperspectral imaging," *ACM Trans. Graph.*, vol. 33, no. 6, pp. 1–11, Nov. 2014.

[12] X. Yuan, T.-H. Tsai, R. Zhu, P. Llull, D. Brady, and L. Carin, "Compressive hyperspectral imaging with side information," *IEEE J. Sel. Topics Signal Process.*, vol. 9, no. 6, pp. 964–976, Sep. 2015.

[13] Z. Meng, M. Qiao, J. Ma, Z. Yu, K. Xu, and X. Yuan, "Snapshot multispectral endomicroscopy," *Opt. Lett.*, vol. 45, no. 14, pp. 3897–3900, 2020.

[14] A. Wagadarikar, R. John, R. Willett, and D. Brady, "Single disperser design for coded aperture snapshot spectral imaging," *Appl. Opt.*, vol. 47, no. 10, pp. B44–B51, 2008.

[15] J. M. Bioucas-Dias and M. A. T. Figueiredo, "A new TwIST: Two-step iterative shrinkage/thresholding algorithms for image restoration," *IEEE Trans. Image Process.*, vol. 16, no. 12, pp. 2992–3004, Dec. 2007.

[16] M. A. T. Figueiredo, R. D. Nowak, and S. J. Wright, "Gradient projection for sparse reconstruction: Application to compressed sensing and other inverse problems," *IEEE J. Sel. Topics Signal Process.*, vol. 1, no. 4, pp. 586–597, Dec. 2007.

[17] X. Yuan, "Generalized alternating projection based total variation minimization for compressive sensing," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2016, pp. 2539–2543.

[18] Y. Liu, X. Yuan, J. Suo, D. J. Brady, and Q. Dai, "Rank minimization for snapshot compressive imaging," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 12, pp. 2990–3006, Dec. 2019.

[19] X. Miao, X. Yuan, Y. Pu, and V. Athitsos, "L-net: Reconstruct hyperspectral images from a snapshot measurement," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2019, pp. 4059–4069.

[20] Y. Cai et al., "Coarse-to-fine sparse transformer for hyperspectral image reconstruction," in *Proc. Eur. Conf. Comput. Vis. (ECCV)* (Lecture Notes in Computer Science). Berlin, Germany: Springer, 2022, pp. 686–704.

[21] Y. Cai et al., "Mask-guided spectral-wise transformer for efficient hyperspectral image reconstruction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 17502–17511.

[22] Z. Cheng et al., "Recurrent neural networks for snapshot compressive imaging," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 2264–2281, Feb. 2023.

[23] Z. Meng, X. Yuan, and S. Jalali, "Deep unfolding for snapshot compressive imaging," *Int. J. Comput. Vis.*, vol. 131, no. 11, pp. 2933–2958, Nov. 2023.

[24] J. Ma, X.-Y. Liu, Z. Shou, and X. Yuan, "Deep tensor ADMM-Net for snapshot compressive imaging," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 10223–10232.

[25] Y. Cai et al., "Degradation-aware unfolding half-shuffle transformer for spectral compressive imaging," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 37749–37761.

[26] X. Zhao, M. Bai, and M. K. Ng, "Nonconvex optimization for robust tensor completion from grossly sparse observations," *J. Sci. Comput.*, vol. 85, no. 2, p. 46, Nov. 2020.

[27] W.-H. Wu, T.-Z. Huang, X.-L. Zhao, J.-L. Wang, and Y.-B. Zheng, "Hyperspectral image denoising via tensor low-rank prior and unsupervised deep spatial–spectral prior," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022, doi: 10.1109/TGRS.2022.3228927.

[28] R. Dian, S. Li, and L. Fang, "Learning a low tensor-train rank representation for hyperspectral image super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 9, pp. 2672–2683, Sep. 2019.

[29] F. L. Hitchcock, "The expression of a tensor or a polyadic as a sum of products," *J. Math. Phys.*, vol. 6, nos. 1–4, pp. 164–189, Apr. 1927.

[30] L. R. Tucker, "Some mathematical notes on three-mode factor analysis," *Psychometrika*, vol. 31, no. 3, pp. 279–311, Sep. 1966.

[31] D. Goldfarb and Z. Qin, "Robust low-rank tensor recovery: Models and algorithms," *SIAM J. Matrix Anal. Appl.*, vol. 35, no. 1, pp. 225–253, Jan. 2014.

[32] W. He, Q. Yao, C. Li, N. Yokoya, and Q. Zhao, "Non-local meets global: An integrated paradigm for hyperspectral denoising," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6861–6870.

[33] W. He et al., "Non-local meets global: An iterative paradigm for hyperspectral image restoration," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 4, pp. 2089–2107, Apr. 2022.

[34] Y. Wang, Y. Han, K. Wang, and X.-L. Zhao, "Total variation regularized nonlocal low-rank tensor train for spectral compressive imaging," *Signal Process.*, vol. 195, Jun. 2022, Art. no. 108464.

[35] D. Kittle, K. Choi, A. Wagadarikar, and D. J. Brady, "Multiframe image estimation for coded aperture snapshot spectral imagers," *Appl. Opt.*, vol. 49, no. 36, pp. 6824–6833, 2010.

[36] L. Wang, Z. Xiong, D. Gao, G. Shi, and F. Wu, "Dual-camera design for coded aperture snapshot spectral imaging," *Appl. Opt.*, vol. 54, no. 4, pp. 848–858, 2015.

[37] L. Wang, Z. Xiong, G. Shi, F. Wu, and W. Zeng, "Adaptive nonlocal sparse representation for dual-camera compressive hyperspectral imaging," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 10, pp. 2104–2111, Oct. 2017.

[38] S. Zhang, L. Wang, Y. Fu, X. Zhong, and H. Huang, "Computational hyperspectral imaging based on dimension-discriminative low-rank tensor recovery," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 10182–10191.

[39] J. Yang et al., "Video compressive sensing using Gaussian mixture models," *IEEE Trans. Image Process.*, vol. 23, no. 11, pp. 4863–4878, Nov. 2014.

[40] J. Yang et al., "Compressive sensing by learning a Gaussian mixture model from measurements," *IEEE Trans. Image Process.*, vol. 24, no. 1, pp. 106–119, Jan. 2015.

[41] Z. Meng, Z. Yu, K. Xu, and X. Yuan, "Self-supervised neural networks for spectral snapshot compressive imaging," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 2602–2611.

[42] H. Qiu, Y. Wang, and D. Meng, "Effective snapshot compressive-spectral imaging via deep denoising and total variation priors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 9123–9132.

[43] Y. Chen, X. Gui, J. Zeng, X.-L. Zhao, and W. He, "Combining low-rank and deep plug-and-play priors for snapshot compressive imaging," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–13, 2023, doi: 10.1109/TNNLS.2023.3294262.

[44] Y. Chen, W. Lai, W. He, X.-L. Zhao, and J. Zeng, "Hyperspectral compressive snapshot reconstruction via coupled low-rank subspace representation and self-supervised deep network," *IEEE Trans. Image Process.*, vol. 33, pp. 926–941, 2024.

[45] P. Xu, L. Liu, Y. Jia, H. Zheng, C. Xu, and L. Xue, "A refinement boosted and attention guided deep FISTA reconstruction framework for compressive spectral imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–12, 2023, doi: 10.1109/TGRS.2023.3257125.

[46] A.-H. Phan, P. Tichavský, and A. Cichocki, "Fast alternating LS algorithms for high order CANDECOMP/PARAFAC tensor factorizations," *IEEE Trans. Signal Process.*, vol. 61, no. 19, pp. 4834–4846, Oct. 2013.

[47] C. Battaglino, G. Ballard, and T. G. Kolda, "A practical randomized CP tensor decomposition," *SIAM J. Matrix Anal. Appl.*, vol. 39, no. 2, pp. 876–901, 2018.

[48] A. Beck and M. Teboulle, "Fast gradient-based algorithms for constrained total variation image denoising and deblurring problems," *IEEE Trans. Image Process.*, vol. 18, no. 11, pp. 2419–2434, Nov. 2009.

[49] I. Choi, D. S. Jeon, G. Nam, D. Gutierrez, and M. H. Kim, "High-quality hyperspectral reconstruction using a spectral prior," *ACM Trans. Graph.*, vol. 36, no. 6, pp. 1–13, Dec. 2017.

[50] B. Arad and O. Ben-Shahar, "Sparse recovery of hyperspectral signal from natural RGB images," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2016, pp. 19–34.

[51] F. Yasuma, T. Mitsunaga, D. Iso, and S. K. Nayar, "Generalized assorted pixel camera: Postcapture control of resolution, dynamic range, and spectrum," *IEEE Trans. Image Process.*, vol. 19, no. 9, pp. 2241–2253, Sep. 2010.

[52] A. Chakrabarti and T. Zickler, "Statistics of real-world hyperspectral images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2011, pp. 193–200.

[53] Z. Meng, J. Ma, and X. Yuan, "End-to-end low cost compressive spectral imaging with spatial–spectral self-attention," in *Proc. Eur. Conf. Comput. Vis. (ECCV)* (Lecture Notes in Computer Science). Berlin, Germany: Springer, 2020, pp. 187–204.

[54] X. Hu et al., "HDNet: High-resolution dual-domain learning for spectral compressive imaging," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Oct. 2022, pp. 17542–17551.

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